# Norman Traffic Case: Domain Summary

Nov 7, 2026 · @Cui Zou

## Page 1: Workflow for each question

All five questions share one data pipeline; they differ only in what they rank, over which period, and where the results are shown. The City wants ranked lists, not project recommendations, built by a process staff can repeat on new data.

**Shared pipeline (done once)**

1. **Combine the five ACOG count files** (FY19, FY21, FY23, FY25, FY26): rename columns to standard names, stack them, turn count text into numbers, keep the highest volume.
2. **Clean road names** into one key per road; fix garbled station names by station ID.
3. **Build one-mile segments**: each road between two neighbouring section-line (grid) roads; keep only arterials and section-line roads.
4. **Add lanes** (from the aerial map), look up **capacity** (ACOG table), compute **V/C**.
5. **Place collisions**: drop private-property crashes, fill missing coordinates, move "From" crashes by direction and distance, match each crash to the segment of the road it happened on.
6. **Assign wards**: each segment to every ward it lies in (boundary roads get both).

**Steps specific to each question**

| Question | What the City asks | Steps after the shared pipeline | Output |
|---|---|---|---|
| Q1 | Top 20 on collisions and traffic volume combined | Keep segments with a volume and at least 1 collision; rank V/C and collisions; score = sum of the two ranks; sort lowest first | Top 20 combined list |
| Q2 | Top 20 on traffic volume | Rank all segments by V/C (highest = 1) | Top 20 V/C list |

| Question | What the City asks | Steps after the shared pipeline | Output |
|---|---|---|---|
| Q3 | Top 20 on collisions | Count collisions per segment (Jan 2016 to Aug 2026); rank (most = 1) | Top 20 collision list |
| Q4 | Top 20 lists for each of the five count years | For each year: that file's volumes only + that fiscal year's crashes; repeat Q1 to Q3; count how many years each segment appears | 15 yearly lists + a persistence table |
| Q5 | Results on the City Council Ward Map with a heat map | Map segments coloured by V/C, collisions and combined rank over the 8 wards; add a crash heat map; count Top 20 segments per ward | Static maps, interactive map, ward representation table |
| Ward lists | V/C, collision and combined lists for each of the 8 wards | Take each ward's segments (boundary segments in both wards) and re-rank them within the ward | 24 ranked lists |

## Page 2: Important assumptions

The rankings rest on the assumptions below. Rows marked **City rule** come from the instructions or the City's written clarifications; the rest are analyst choices that are defensible if stated in the notebook.

| Area | Assumption | Why |
|---|---|---|
| Volume | If a location was counted more than once (several stations, years, or two counts in one row), use the **highest** volume | City rule (clarification Q1) |
| Volume | Where to read it: FY19 = column L and the text count in column N; FY21 = column N and the text in column Q; FY23 and FY25 = sum of the directions in column L; FY26 = sum of column J | City rule (clarification Q1) |
| Volume | The two count rows with no station ID or coordinates (end of FY25 and FY26) are left out | Cannot be placed on a road; both are lower than other counts on those roads |
| Road names | Garbled FY23 names (e.g. "(E Lindsey St)12th Ave NW") are corrected by station ID | Station IDs stay the same across years |
| Segments | A segment is one road between two neighbouring one-mile grid roads (about 1 mile) | Matches the City's examples (Rock Creek Rd from 12th to 24th Ave NE); objective and repeatable |
| Road classes | Keep Principal Arterials, Minor Arterials and section-line roads; drop I-35, SH-9, collectors and local streets | City rule (Jargon sheet) |
| Road classes | Daws St dropped by hand; Boyd St (Collector, up to about 18,000 vpd) dropped by rule | Daws is labelled Minor Arterial but carries about 800 vpd; including Boyd is a defensible alternative if stated |
| Collisions | A crash counts for the road it happened **on** (StreetName), not the cross street | City rule (the 48th Ave NE / Franklin Rd example) |
| Collisions | "From" crashes are moved Distance feet in Direction from the intersection; distances over 1 mile (e.g. 9999) are treated as unknown | Lat/Lon is the nearest intersection, not the crash |

| Area | Assumption | Why |
|---|---|---|
| Collisions | A crash within about 15 m of a grid intersection counts for both adjacent segments of that road | The intersection ends both segments |
| Collisions | Private-property (parking lot) crashes and crashes with no usable location are dropped | Not on a public road segment |
| Collisions | Crashes on roads or miles with no traffic count are not used | City rule (the "major hint") |
| Collisions | Raw counts over Jan 2016 to Aug 2026, not rates | What the City asked for; 2026 is a partial year |
| Lanes | Lanes are counted in both directions from aerial imagery; the provided lanes file holds **estimates** to verify | City rule (count from the interactive map); V/C depends heavily on lanes |
| Capacity | ACOG LOS E capacities (2025 OCARTS Plan); a 3-lane road uses the 2-lane value; one-way = 11,000 vpd per lane | Newest ACOG standard; ACOG has no 3-lane row |
| Wards | A segment belongs to every ward holding at least 10% of it, so boundary segments appear in both lists | City rule (count in both wards) |
| Ranking | Ties share the best rank (1, 2, 2, 4); highest V/C or most collisions = rank 1 | Standard competition ranking |
| Ranking | Combined ranking uses only segments with a volume and at least 1 collision; ranks are recomputed within that set (and within each ward); ties broken by higher V/C | City rule (both data needed); tie-break is an analyst choice |
| Q4 years | Each count year uses that file's volumes and crashes from the matching fiscal year (July to June); only segments counted that year enter its V/C and combined lists | Count years are fiscal years and are not consecutive |

# Page 3: Important metrics

Two numbers drive every ranking: the **V/C ratio** (how full the road is) and the **collision count** (how many crashes happen on it). The rest of the metrics below feed into one of those two.

**V/C = highest daily volume (vpd) ÷ daily capacity (vpd)**

**Combined score = V/C rank + collision rank** (lowest score = highest priority)

| Metric | What it means | How it is computed | Example |
|---|---|---|---|
| Vehicles per day (vpd) | Two-way daily traffic on a road | Add the directional counts in the count text; ignore speeds (mph) and dates | NB 2,009 + SB 1,889 = 3,898 vpd |
| Highest volume | The volume used for a segment | Largest value across all count records and years on that segment | 18,000 (FY19) and 20,000 (FY25) → 20,000 |
| Lanes | Through lanes in both directions, plus turn lanes and medians | Read from aerial imagery; coded 2, 3, 4U, 4D, 5, 6U, 6D, 1W2 | 4D = two lanes each way with a median |
| Level of service (LOS) | A grade from A (free flow) to F (breakdown) for how well traffic moves | Sets which capacity column to use: LOS C (2020 plan) or LOS E (2025 plan) | LOS E capacities are higher, so V/C is lower; the order barely changes |
| Capacity | Most vehicles a road can carry per day at that level of service | Look up the lane code in the ACOG table | 2-lane = 17,100; 4-lane divided = 38,000; 5-lane = 36,000 (LOS E) |
| V/C ratio | How full the road is: 1.0 = at capacity, above 1.0 = over capacity | Highest volume ÷ capacity | 12th Ave NE/SE (Main to Robinson): 31,629 ÷ 38,000 = 0.83 |

| Metric | What it means | How it is computed | Example |
|---|---|---|---|
| V/C level | Plain-language band used in the tutorials (not an official standard) | Under 0.50 low; 0.50 to 0.75 moderate; 0.75 to 1.0 near capacity; over 1.0 over capacity | 0.83 = near capacity |
| Collisions per segment | Crashes that happened on that road within that mile | Count of crashes matched to the segment, Jan 2016 to Aug 2026 | Main St (36th to 24th Ave NW): about 680 crashes, the most in the study |
| Rank (with ties) | Position in a sorted list | Largest value = 1; equal values share the best rank | Values 50, 40, 40, 30 → ranks 1, 2, 2, 4 |
| Combined score | Congestion and safety together | V/C rank + collision rank, among segments with both; lowest first | 12th Ave NE/SE (Main to Robinson): 3rd in V/C + 2nd in collisions = 5 → #1 overall |
| Fiscal year (FY) | The year label of a count file | July 1 to June 30, named by the year it ends | FY19 = July 2018 to June 2019 |
| Persistence | Whether a problem repeats | Number of the five count years a segment appears in a Top 20 | Flood Ave (Robinson to Rock Creek): combined Top 20 in all 5 years |
| Primary cause | What caused the crash | CauseCode in Base_Data (for Unit 1, the vehicle at fault), decoded with Code_Book.xlsx | -67239 = Changed Lanes Unsafely |